SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Xeno Force
Project / Solution Name	ArcTalent
Name (Leader)	Emmanuel Heng Zi Teng
Email & Phone (Leader)	enuelhzt1524@gmail.com +6010-271-6363
University Affiliation(s)	Universiti Malaya; Universiti Teknologi Malaysia; Management and Science University; Swinburne University of Technology.
Live Prototype URL	https://arc-talent--emmanuelheng.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/MantouOP/ArcTalent

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

ArcTalent is a full-stack Next.js application hosted on Replit. Its React and TypeScript interface collects learner profiles, career goals, learning constraints and workforce-planning inputs.

A shared TypeScript engine calculates career suitability, skill gaps, scenario comparisons and course schedules. This engine powers both the interface and backend analysis, keeping calculations consistent.

Next.js API routes handle PDF extraction, AI requests, public evidence refreshes and PDF report generation. Optional generative AI is accessed through the Vercel AI SDK using RunPod or a configured OpenAI-compatible gateway. It supports CV skill suggestions, career-goal analysis, workforce-plan drafting, chat and study guidance. Zod validates structured inputs and outputs, while supported features provide local fallbacks.

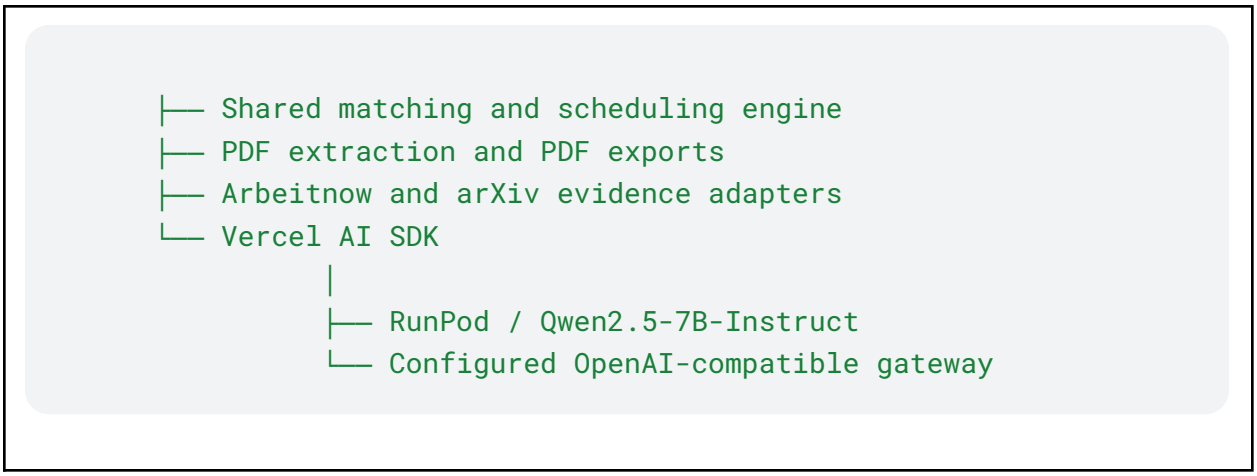
Profiles, preferences, progress, planning choices and pictures are stored as typed JSON in browser localStorage. The prototype does not use a persistent cloud database or authentication service.

Public evidence adapters retrieve bounded samples from Arbeitnow and arXiv. Curated course records supply canonical titles and direct links.

Users review suggestions, edit proposed plans and explicitly confirm skill-level changes. AI-generated text does not independently determine employment decisions.

Optional diagram:

```
None
User interface
      |
      ├── Browser localStorage
      |   Profiles, preferences, progress and pictures
      |
      ▼
Next.js backend on Replit
      |
```



2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	***Purpose in Project***	***License***
TypeScript 7.0.2	*Typed application logic, shared data structures and backend implementation*	*Apache-2.0*
JavaScript, HTML and CSS	*Browser behaviour, document structure and interface styling*	*Language standards; no separate library licence*
Next.js 16.4.2	*Full-stack framework, application routing and backend API routes*	*MIT*
React 19.3.1 and React DOM 19.3.1	*Interface components, rendering and application state*	*MIT*
Zod 4.6.7	*Validation of API requests, AI responses and saved application state*	*MIT*

Vercel AI SDK — `ai` 7.0.148	*Model communication and structured AI output*	*Apache-2.0*
`@ai-sdk/openai-compatible` 3.0.75	*Connection to compatible AI gateways and RunPod endpoints*	*Apache-2.0*
`pdf-parse` 2.4.5	*Text extraction from uploaded text-based PDFs*	*Apache-2.0*
PDFKit 0.20.2	*Searchable personal and workforce PDF reports*	*MIT*
Lucide React 1.61.0	*Interface and navigation icons*	*ISC*
`tsx` 4.23.15	*Running TypeScript tests and benchmark scripts*	*MIT*
Prettier 3.9.10	*Source-code and documentation formatting*	*MIT*
Node.js — minimum 22; local development verified with 24.13.1	*Server runtime and development tooling*	*MIT, with bundled component notices*
Noto Sans SC, bundled font	*English, Malay and Chinese text in exported PDFs*	*SIL Open Font License 1.1*
Inter and Outfit, loaded through Google Fonts	*Website typography*	*SIL Open Font License 1.1 — Inter licence, Outfit licence*

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier
RunPod OpenAI-compatible inference endpoint	Calls to the configured Qwen model for supported AI functions, including CV extraction and career-goal analysis	Sponsor-supported use recorded in the earlier declaration; **[CONFIRM account billing/credit tier]**
Configured OpenAI-compatible / LiteLLM gateway	Alternative inference provider for career guidance, study coaching and workforce-plan drafting; documented verification used `gpt-5.4-mini`	**[CONFIRM sponsor-provided or paid access, and official gateway provider name]**
Arbeitnow Job Board API — `https://www.arbeitnow.com/api/job-board-api`	Bounded public job-posting sample used to derive skill-mention signals	Free public API
arXiv cs.AI RSS — `https://rss.arxiv.org/rss/cs.AI`	Recent AI research metadata used as contextual evidence	Free public feed
Google Fonts CSS/font delivery	Loads Inter and Outfit for the interface	Free public font delivery
Course-provider image servers and YouTube thumbnail delivery	Loads course covers and companion-video thumbnails	Public web resources; subject to provider terms
Unicorn Studio scene delivery — historical trial only	Loaded the supplied decorative scene during an earlier prototype; subsequently removed	**[CONFIRM historical account tier]**; not used by the current application

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	***Purpose in Project***	***Plan / Region***
Replit	*Hosts the Next.js application, backend routes, build process, Secrets and public deployment*	*Autoscale; Asia region recorded in deployment verification.* ***[CONFIRM account/credit plan]***
RunPod	*GPU inference endpoint for the configured Qwen model*	*Serverless endpoint;* ***[CONFIRM account tier and endpoint region]***
OpenAI-compatible AI gateway	*Hosts or routes inference to the configured hosted model*	***[CONFIRM provider, access plan and processing region]***
GitHub	*Public source repository, incremental history and GitHub Actions verification*	*Public repository and Actions; managed GitHub service*
Browser localStorage	*Saves profiles, preferences, progress, workforce settings and pictures*	*Local device storage; not cloud storage*

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen/Qwen2.5-7B-Instruct	Qwen / Alibaba; accessed through the team's RunPod endpoint	Pre-trained model used for structured inference; no team fine-tuning or training	Apache License 2.0, plus hosting-service terms
`gpt-5.4-mini`, as identified in the documented gateway verification	Configured OpenAI-compatible gateway	Pre-trained hosted model used for supported AI guidance and planning functions	Applicable model-provider and gateway terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	*Source / URL*	*Type*	*License / Terms*	*Personal Data?*
Eight fictional worker profiles	*Project fixture in* `src/lib/catalogue.ts`	*Synthetic*	*Project-created fixture under Apache-2.0*	*N — fictional profiles*
Synthetic CV samples	`public/synthetic-cv.txt` *and* `public/synthetic-cv.pdf`	*Synthetic*	*Project-created fixture under Apache-2.0*	*N — fabricated demonstration data*

Baseline and updated evidence snapshots	*Project fixtures in* `src/lib/catalogue.ts`	*Synthetic*	*Project-created fixture under Apache-2.0*	*N*
Curated course catalogue	*OpenLearn, OpenClassrooms, Microsoft Learn, SQLBolt and MDN; links recorded in the source register*	*Public / self-collected metadata*	*Provider terms; course content and images retain their own rights*	*N in intended catalogue fields*
Occupational reference material	*O*NET*	*Public*	*Applicable O*NET terms and attribution requirements; project skill requirements and exposure factors are separate assumptions*	*N in intended occupational fields*
Job-posting sample	*Arbeitnow API*	*Public*	*Provider API and content terms*	*Y, potentially — upstream descriptions may contain contact details*
AI research metadata	*arXiv cs.AI RSS*	*Public*	*Descriptive metadata is CC0 under arXiv API terms; papers retain separate rights*	*Y, potentially — public author metadata may identify people*
Training-cost and wage references	*HRD Corp Allowable Cost Matrix and DOSM wage statistics, linked in the source register*	*Public aggregate references*	*Respective publication terms and attribution*	*N — aggregate figures*

SDG definitions and target references	*United Nations SDGs*	*Public reference material*	*UN website terms; contextual project mappings are identified as interpretations*	*N*
User-entered profiles, CVs, goals, reflections and workforce reports	*Entered or uploaded by the user*	*Private / user-provided runtime input*	*Used for the requested processing; not an openly licensed dataset*	*Y — may contain personal or sensitive information*

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
OpenAI Codex / ChatGPT — versions varied	Ideation, architecture, implementation, interface changes, debugging, tests, research assistance and documentation	Major
Google Antigravity	Development assistance, debugging and project refinement	Moderate
Claude	Project explanations, content refinement and supporting development work	Moderate
GitHub Copilot	Code completion and assistance with implementation and tests	Moderate
Superpowers and Taste Skill workflows within Codex — installed repository versions	Planning, debugging, review and interface-development guidance	Moderate; workflow extensions rather than separate models

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-generated code and text were treated as draft material requiring verification. Source review, TypeScript checks, automated tests and browser checks were used to assess behaviour and identify defects. A documented Replit release passed 250 unit tests, a production build and 39 integration checks. AI suggestions were checked against application schemas and canonical course records; CV evidence quotes were checked against the supplied text. The team remains responsible for reviewing the final implementation, validating assumptions, correcting unsupported claims and ensuring that the submission accurately describes its own contributions and use of AI.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Cloud development and deployment platform	Hosted the full-stack application and backend routes, managed server-side configuration and published the working prototype
RunPod	GPU inference platform and endpoint	Provided access to the configured Qwen2.5-7B-Instruct model for structured AI inference
Aiversecity / Domain 1 challenge provider	Workforce challenge brief and domain requirements	Guided the focus on transferable skills, changing AI-adoption scenarios and adaptive reskilling pathways
	OpenAI-compatible GenAI gateway access	Provided the alternative inference connection used by the documented workforce-planning verification

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides the current public hosting environment, while RunPod and the configured gateway supply optional generative-AI inference. These services supported the prototype's implementation and demonstration, but the architecture is replaceable. The matching engine, scheduler and cost calculations do not require a generative model. Supported AI features provide local fallbacks, and the provider adapter can connect to another compatible endpoint. The application could be deployed on another suitable Node.js host, subject to configuration and testing. Sponsor resources therefore support delivery without making the core planning logic dependent on one model provider.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

ArcTalent accepts profile details, skills, interests, learning constraints, pictures and optional CVs, reflections and workforce reports. Saved planning data remains in browser localStorage until reset or browser-data removal; it has no automatic expiry. Chat and unsaved drafts are transient.

Text PDFs are processed by the application server. ArcTalent does not persist raw CV files or text in an application database. Requested AI features send relevant inputs to the configured provider, including reviewed CV text when AI extraction is selected. The workforce planner masks common email, phone and identity-number patterns, but this does not guarantee complete anonymisation.

Requests use HTTPS, and API credentials are configured server-side. Third-party fonts and images also generate ordinary web requests. Hosting and AI-provider logging or retention follow their own policies; provider-side zero retention has not been verified. The prototype has no multi-user authentication or employer access-control system.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Generative models can suggest unsupported skills or unsuitable plans. ArcTalent mitigates this through structured validation, source-quote checks for CV candidates, canonical course links and user review. An exact quote does not independently prove proficiency.

Matching weights, role-exposure factors and course-to-skill mappings are project assumptions. Scores are planning aids rather than hiring probabilities. O*NET provides US occupational context, while the public job adapter samples European postings; Malaysian relevance requires further validation.

AI services and public feeds may time out, be unavailable or impose usage limits. The application uses bounded requests, timeouts, labelled local fallbacks and simulated replay data. Fallback output may be less detailed.

The PDF parser requires readable text and does not provide OCR. Pattern-based identifier masking is incomplete protection for sensitive reports. Practice badges record self-study activity, not independent certification. Employment, productivity and SDG outcomes have not been measured.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

☐ - stands for ✅

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]
1.11.2026